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Enantioselective difluoromethylation of unactivated alkyl halides via a formal nickel(II/IV) cycle

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Nickel-catalysed enantioconvergent cross-coupling has emerged as a key strategy for constructing stereochemically defined C(sp^3) centres. However, the reliance on nickel(I)/nickel(III) catalytic cycles has limited the range of compatible alkyl electrophiles and nucleophiles. Here we report an aryl-radical-enabled oxidative addition mechanism that connects Ni^{II} and formal Ni^{IV} oxidation states, enabling the enantioselective difluoromethylation of unactivated alkyl halides. Mechanistic investigations, including X-ray crystallography, electron paramagnetic resonance spectroscopy and density functional theory calculations, support the involvement of a nickel(II/III/IV) cycle via the stepwise oxidative addition of Ni^{II} intermediates. The broad substrate scope and compatibility with pharmaceutically relevant molecules highlight the potential of this catalytic platform in asymmetric catalysis.

Driven by the high demand for enantiomerically pure pharmaceuticals, agrochemicals and materials, the development of asymmetric transformations continues to be a cornerstone of modern synthetic chemistry[1–4]. In this context, transition-metal-catalysed asymmetric reactions have enabled a broad range of enantioselective carbon–carbon and carbon–heteroatom bond formations[5]. Among these reactions, nickel-catalysed enantioconvergent cross-coupling of alkyl electrophiles with nucleophiles has emerged as a powerful strategy for the construction of stereochemically defined C(sp^3) centres[6–12]. These reactions are mainly governed by two well-established mechanistic pathways, a $Ni^{I}/Ni^{II}/Ni^{III}$ cycle and a $Ni^{0}/Ni^{II}/Ni^{III}$ cycle, with the latter being more prevalent in asymmetric cross-electrophile couplings[8,13] (Fig. 1a). Each of these Ni^{III}-based pathways has successfully engaged a wide range of coupling partners for asymmetric bond formation. Nonetheless, the scope of these pathways has been largely restricted to electron-rich nucleophiles, leaving electron-deficient nucleophiles, such as fluoroalkyl groups, relatively unexplored. Furthermore, with few exceptions, current methods are limited to activated alkyl electrophiles[14–16]. Therefore, a necessary objective for the continued advancement of asymmetric metal catalysis is the development of distinct catalytic activation modes that enable previously inaccessible transformations.

High-valent organo-Ni^{IV} complexes, while rare, have been increasingly investigated, with recent reports describing well-defined examples characterized by X-ray diffraction[17,18] (Fig. 1b). Stoichiometric organometallic studies suggest that such Ni^{IV} complexes can be generated from Ni^{II} species via chemical oxidation[19]. Pioneering studies by Sanford[20–22], Mirica[23–25] and others have demonstrated that these alkyl–Ni^{IV} complexes can exhibit catalytically relevant reactivity distinct from their Ni^{III} counterparts. While a few elegant studies have demonstrated the feasibility of Ni^{IV}-mediated catalysis using specially designed substrates[26,27], harnessing the reactivity of Ni^{IV} intermediates in asymmetric bond-forming reactions remains largely unexplored.

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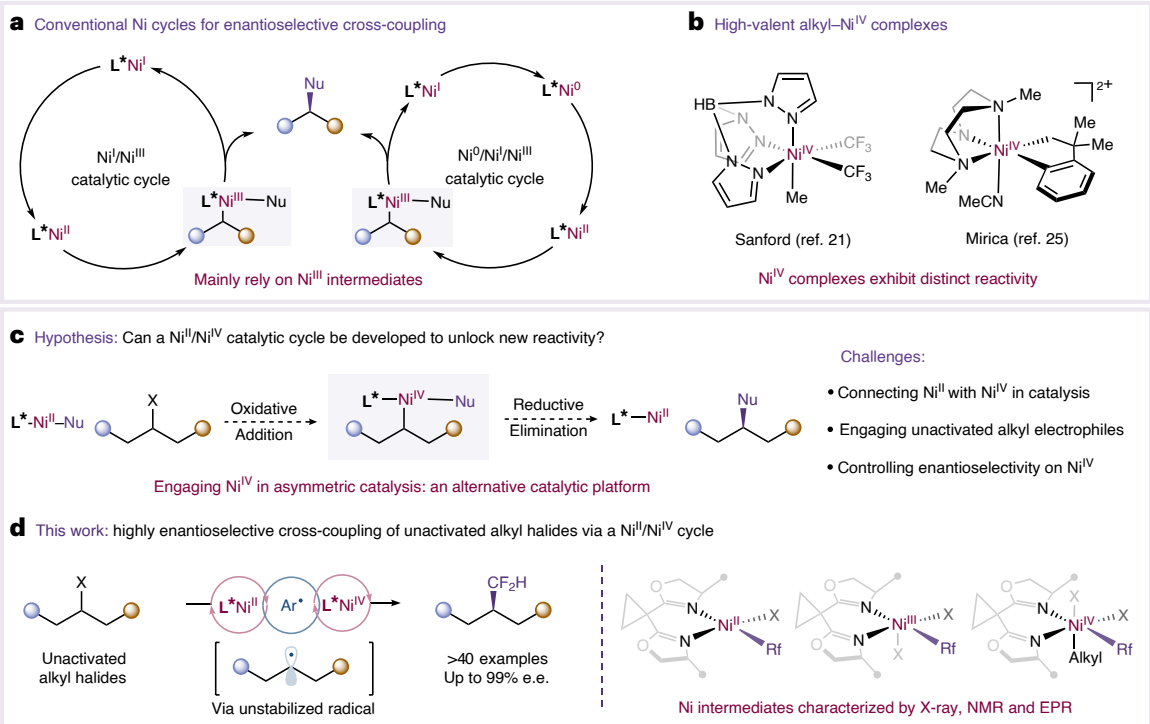

Fig. 1 | Development of catalytic platform for asymmetric synthesis via a Ni^{II}/Ni^{IV} cycle. a, Conventional nickel catalytic cycles rely mainly on Ni^{I}/Ni^{III} cycles. **b**, Alkyl–Ni^{IV} complexes have been isolated, albeit with limited relevance to catalysis. Me, methyl. **c**, A platform based on a Ni^{II}/Ni^{IV} cycle. **d**, This work reports a well-defined Ni^{II}/Ni^{IV} cycle for enantioselective difluoromethylation of unactivated alkyl iodides.

In particular, enantioselective cross-coupling via a formal Ni^{II}/Ni^{IV} redox cycle remains elusive, yet highly intriguing.

We recently questioned whether a catalytic cycle involving the formation of Ni^{IV} intermediates via oxidative addition of Ni^{II} precursors could enable the development of cross-coupling reactions, ideally in an enantioselective manner that can engage unactivated alkyl electrophiles (Fig. 1c). This approach is especially intriguing given that the reductive elimination of highly electronegative groups such as C–fluorine[28] and C–fluoroalkyl[20] bonds is more favourable at Ni^{IV} centres than at Ni^{III} centres. Moreover, the increased electrophilicity of the Ni^{IV} complexes could potentially enhance interactions between the metal centres and the substrate binding sites, thereby improving enantio-discrimination between two alkyl groups. Consequently, this catalytic pathway is expected to complement the existing Ni^{III}-based pathways and expand the scope of asymmetric nickel catalysis.

Organic molecules bearing difluoromethyl (CF_2H) groups at stereogenic centres have attracted growing interest in the pharmaceutical industry. The unique properties of the CF_2H group stem from its ability to act as a lipophilic hydrogen bond donor, rendering it a valuable bioisostere of hydroxyl, amine and thiol groups[29–33]. Several US Food and Drug Administration-approved pharmaceutical molecules have demonstrated a critical role of stereochemically defined CF_2H groups in modulating biological activity through hydrogen-bonding interaction with protein active sites[34,35]. Despite this promise, broader exploration of pharmaceutical candidates bearing CF_2H groups at chiral centres has been hindered by the challenges associated with their enantioselective installation[36–39]. Among synthetic strategies, enantioconvergent difluoromethylation of alkyl electrophiles represents one of the most attractive yet least developed approaches, with the few existing methods largely restricted to activated alkyl electrophiles[40,41]. Therefore, a mechanistic pathway that can engage nucleophilic difluoromethyl reagents and unactivated alkyl halides in asymmetric catalysis could enable a mild and broadly applicable strategy towards the enantioselective difluoromethylation.

In this study, we report an aryl-radical-enabled oxidative addition pathway that connects Ni^{II} and Ni^{IV} oxidation states in catalysis while simultaneously engaging unactivated alkyl halides in asymmetric cross-couplings. This catalytic strategy has been exploited to deliver a broadly applicable approach to the enantioselective difluoromethylation of unactivated alkyl iodides (Fig. 1d). Mechanistic studies, including nuclear magnetic resonance (NMR) spectroscopy, electron paramagnetic resonance (EPR) spectroscopy, X-ray crystallography and density functional theory (DFT) calculations, support the involvement of a formal Ni^{II}/Ni^{IV} cycle in this transformation.

Results

Reaction development

Our proposed mechanism for the enantioselective difluoromethylation via a $Ni^{II/IV}$ cycle is outlined in Fig. 2a. Transmetallation of a Ni^{II} catalyst **1** with a nucleophilic CF_2H reagent should produce a [Ni^{II}–CF_2H] intermediate **2**. We hypothesize that a suitable aryl-based oxidant **3** should undergo a single electron transfer (SET) with **2** to yield a Ni^{III} intermediate **4** and an aryl radical **5** (ref. 27). The subsequent iodine abstraction from an alkyl iodide **6** by the aryl radical, with a bimolecular rate of 10^9 M^{-1} s^{-1}, would then yield an unstabilized alkyl radical **7** (ref. 42). At this time, we believe that **7** would undergo rapid trapping by the Ni^{III} complex **4** to yield an alkyl–Ni^{IV} intermediate **8** (ref. 21). We expect that this trapping step could be facilitated by a weakly coordinating group within the substrate that interacts with the Ni^{IV} centre. Notably, the feasibility of this aryl-radical-enabled oxidative addition of Ni^{II} to Ni^{IV} has been indicated in Sanford's recent work with well-defined nickel complexes[27]. Reductive elimination from this high-valent Ni^{IV} intermediate would yield the desired product **9** and simultaneously regenerate the Ni^{II} catalyst **1**. We reason that, with the aid of a chiral ligand, the CF_2H product would be generated in an

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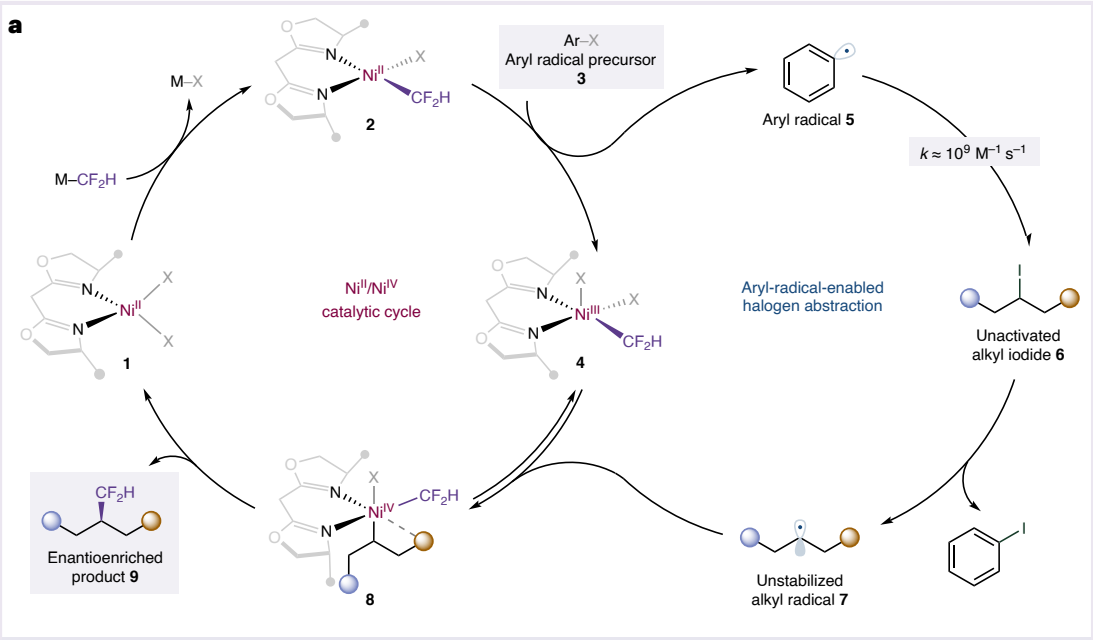

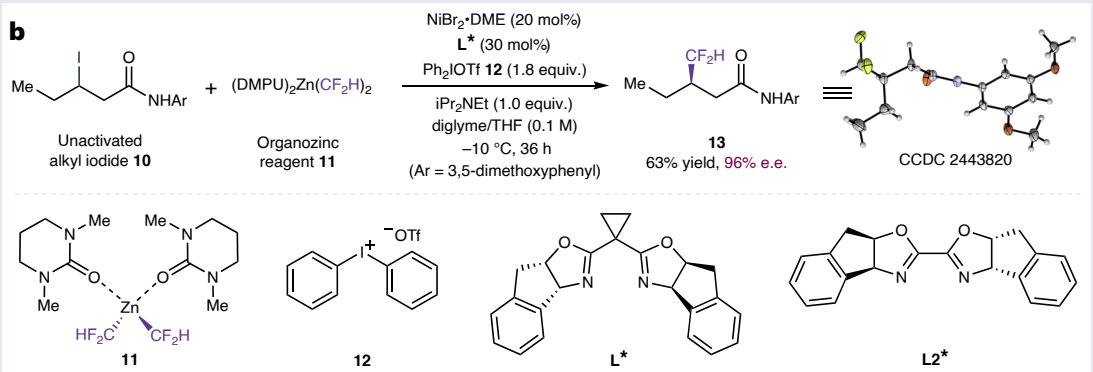

Entry	Variation from standard conditions	Yield[a]	e.e.	Entry	Variation from standard conditions	Yield[a]	e.e.
1	None	63% (55%)[b]	96%	6	Mes_2IOTf (Mes = mesityl)	51%	94%
2	$NiBr_2$ instead of $NiBr_2$•DME	52%	96%	7	Diazonium salt instead of **12**	<1%	N.D.
3	$Cu(CH_3CN)_4PF_6$	19%	0%	8	No iPr_2NEt	51%	96%
4	**L2*** instead of **L***	59%	–33%	9	No $NiBr_2$•DME	<1%	N.D.
5	No **L***	5%	N.D.	10	No PhI_2OTf	<1%	N.D.

Fig. 2 | Development of enantioselective difluoromethylation of alkyl iodides. a, Proposed Ni^{II}/Ni^{IV} catalytic cycle for enantioselective difluoromethylation of alkyl iodides. **b**, Reaction development and optimization. [a]Yield determined by ^{19}F NMR at 0.05 mmol scale. [b]Isolated yield at 0.2 mmol scale. iPr, isopropyl; N.D., not determined.

enantioselective manner. Alternatively, the C–CF_2H bond-forming step could proceed via a formal Ni^{IV} pathway. In this case, the alkyl radical could react with the Ni^{III} intermediate via an S_H2 mechanism, a pathway that also remains elusive for achieving asymmetric bond formation[43].

Extensive optimization of reaction conditions revealed that a combination of $NiBr_2$·DME (where DME is dimethoxyethane) and a commercially available ligand **L*** could convert an alkyl iodide **10**, in the presence of $(DMPU)_2Zn(CF_2H)_2$ **11** (where DMPU is *N*,*N*′-dimethylpropyleneurea) and a diaryliodinium salt **12**, to the desired product **13** in 63% yield and with 96% enantiomeric excess (e.e.) (Fig. 2b, entry **1**). The use of the zinc-based CF_2H reagent **11** was crucial for this transformation, probably due to its ability to undergo efficient transmetallation with the Ni^{II} catalysts[44]. The use of other nickel(II) salts could also promote the reaction with the same enantioselectivity (entry **2**), although the use of copper catalysts was essentially ineffective (entry **3**). The commercially available ligand **L*** is vital for this transformation, and other ligands, such as **L2***, were less effective (entries **4** and **5**). Different diaryliodonium salts afforded the desired product in similar enantioselectivity (entry **6**), although the analogous arenediazonium salts were ineffective (entry **7**), in part due to the poor solubility of the latter in the reaction system. Notably, while diisopropylethyl amine (iPr_2NEt) was found to increase the yield of the reaction, the desired product was also formed with high enantioselectivity without this additive (entry **8**). We reason that the tertiary amine might form an electron donor–acceptor complex with the diaryliodonium salt, which facilitates the SET with the Ni^{II} intermediate[45]. Importantly, no reactivity was observed when the reactions were conducted in the absence of either the nickel catalyst or the diaryliodonium salt (entries **9** and **10**). We expect that the β-carbonyl group of the substrate serves as a weakly coordinating group to the nickel centre, directing this enantioselective difluoromethylation reaction.

Mechanistic investigation

The unusual catalytic system and the high enantioselectivity of the product prompted us to conduct mechanistic studies to elucidate this asymmetric difluoromethylation reaction (Fig. 3). Our first objective was to determine whether a catalytically relevant [Ni^{II}–CF_2H] species could transfer CF_2H groups under the diaryliodonium-promoted conditions (Fig. 3a). Interestingly, mixing (dtbbpy)$NiBr_2$ **14** (dtbbpy = 4,4′-di-*tert*-butyl-2,2′-bipyridine) with $(DMPU)_2Zn(CF_2H)_2$ resulted in the formation of a diamagnetic species, (dtbbpy)$Ni^{II}Br(CF_2H)$ **15**, confirmed by NMR spectroscopy and X-ray crystallography. While **15** could not directly react with alkyl iodide **10**, the addition of a diaryliodonium salt resulted in the disappearance of **15** and the concurrent formation of the product **13** in 42% yield. The analogous [Ni^{II}–CF_2H] species **16** bound with the chiral ligand **L*** was also formed via the transmetallation process and was found to react with **10** under diaryliodonium-enabled conditions, affording **13** in high enantioselectivity (93% e.e.). By contrast, Ni^{II} complex **17**, derived from a less effective chiral ligand $\mathbf{L_2}$*, produced the product with only 21% e.e, aligning well with the poor enantioselectivity observed under catalytic conditions (−33% e.e., at −10 °C). These findings strongly support **L***$Ni^{II}Br(CF_2H)$ species as key reactive intermediates capable of transferring the CF_2H groups under diaryliodonium-promoted conditions.

To further investigate the proposed Ni^{III} intermediates and aryl radicals, we used EPR spectroscopy (Fig. 3b). These candidate species—Ni^{I}, low-spin Ni^{III} and the aryl radical—all possess an $S = ½$ ground state, making them detectable by EPR. By contrast, the Ni^{II} precatalyst, which has a d^8 electronic configuration, is expected to be EPR-silent ($S = 0$ or $S = 1$) under perpendicular-mode conditions. As expected, mixing $NiBr_2$·DME and chiral ligand **L***, generating **L***$Ni^{II}Br_2$ **18** precatalyst, yielded no observable EPR signals (Supplementary Fig. 18). Although alkyl metal reagents are known to reduce Ni^{II} to Ni^{I} in situ[46], mixing **L***$NiBr_2$ with $(DMPU)_2Zn(CF_2H)_2$ did not produce detectable paramagnetic signals (Supplementary Fig. 18). These results indicate that no Ni^{I} species were generated under these conditions, and that $(DMPU)_2Zn(CF_2H)_2$ transfers the CF_2H group to the Ni^{II} precatalyst to generate **L***$Ni^{II}Br(CF_2H)$ **16** via transmetallation, consistent with observations from NMR studies.

Upon addition of a diaryliodonium salt to **L***$Ni^{II}Br(CF_2H)$ **16**, after a 15-s incubation at −35 °C, paramagnetic EPR signals appeared (Fig. 3b(i)). Simulation of the spectrum revealed an $S = ½$ species with slightly rhombic g values of 2.30, 2.23 and 2.01 (Fig. 3b(i) and Supplementary Fig. 19), consistent with a low-spin Ni^{III} species possessing a dz^2 ground state[21,25,47,48]. This supports our hypothesis that the Ni^{II} complex undergoes SET to the diaryliodonium reagent, forming a Ni^{III} intermediate.

In addition to the Ni^{III} EPR signals, a sharp signal at $g \approx 2.0$ (3,380 G) was observed, probably corresponding to an aryl radical formed via SET to the diaryliodonium reagent. To confirm its identity, we performed radical-trapping experiments using 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO). Upon addition of the diaryliodonium salt to **L***$Ni^{II}Br(CF_2H)$ **16** in the presence of DMPO, the resulting EPR spectrum displayed both the characteristic $S = ½$ low-spin Ni^{III} signal and a series of hyperfine around $g \approx 2.0$ (Fig. 3b(ii)). These hyperfine features could arise either from superhyperfine interactions associated with the Ni^{III} complex or from organic radical species. To distinguish between these possibilities, we leveraged the well-established difference in spin–lattice relaxation times between metal-centred and organic radicals. If the sharp peaks stemmed from superhyperfine interactions, their relative intensity would remain unchanged across temperatures. By contrast, signals from organic radicals would exhibit temperature-dependent changes in intensity. EPR spectra collected at two distinct temperatures revealed such temperature-dependent variations in the relative intensity of the sharp peaks compared to the Ni^{III} signal (Supplementary Fig. 20), supporting their assignment to an organic radical. Spectral simulation further confirmed the presence of a broader Ni^{III} signal (g = 2.30, 2.19 and 2.01) and an organic radical species (g = 2.008, 2.008 and 2.002), which exhibited hyperfine coupling to both nitrogen (A_N = 7 G, 7 G and 27 G) and hydrogen nuclei (A_H = 27 G, 16 G and 29 G). These features are consistent with the previously reported DMPO–aryl radical adduct **19** (refs. 41,49), the formation of which has also been confirmed by high-resolution mass spectrometry (Supplementary Fig. 12). Altogether, these EPR studies confirm the formation of both Ni^{III} and aryl radical species via SET from the Ni^{II} complex to the diaryliodonium salt.

Similar EPR characterization was also carried out on (dtbbpy)$Ni^{II}Br(CF_2H)$ **15**. Upon reaction with the diaryliodonium salt, two distinct Ni^{III} species were detected, with the major signals exhibiting g values of 2.30, 2.23 and 2.01, and 2.27, 2.18 and 2.01, respectively (Fig. 3b(iii) and Supplementary Figs. 21–23). These values closely resemble those observed for the Ni^{III} species formed from **L***$Ni^{II}Br(CF_2H)$ **16**. When the reaction was performed in the presence of DMPO, a similar DMPO-trapped aryl radical adduct was observed (Fig. 3b(iv)). These results support a common mechanistic pathway for both ligand systems, in which SET from the Ni^{II} complex to the diaryliodonium salt generates both a Ni^{III} intermediate and an aryl radical.

The feasibility of forming fluoroalkyl–Ni^{IV} species via the aryl-radical-enabled oxidative addition of Ni^{II} precursors was then investigated. Although our attempts to observe [Ni^{IV}–CF_2H] species were unsuccessful presumably due to their high reactivity, the studies with related trifluoromethyl (CF_3) complexes provide strong evidence for this pathway (Fig. 3c). Thus, oxidation of (dtbbpy)$Ni^{II}(CF_3)_2$ **20**, with ethyl iodide **21** and a diazonium salt **22** led to the detection of an unstable alkyl–Ni^{IV} complex **23** as detected by ^{19}F NMR. No reaction occurred in the absence of the diazonium salt. Furthermore, the reaction of an analogous trifluoromethyl nickel complex **24** bound with a trispyrazolylborate (Tp) ligand—known for stabilizing high-valent Ni species—with alkyl iodide **25** and a diaryliodonium salt **26** enabled the isolation of an alkyl–Ni^{IV} complex **27** via this aryl-radical-enabled oxidative addition. The analogous reactivity has also been observed by Sanford using an arenediazonium salt as the aryl radical precursor[20]. Notably, these primary alkyl iodides are competent substrates of the difluoromethylation reactions (Supplementary Fig. 18). Collectively, these mechanistic studies provide strong evidence for an unusual $Ni^{II/IV}$ cycle in the asymmetric difluoromethylation reaction.

Substrate scope and synthetic utility

The generality of this asymmetric difluoromethylation reaction was then investigated (Fig. 4). A group of β-iodoamides could be selectively converted to the desired products with high enantioselectivity (Fig. 4a). The length or the steric hindrance of the alkyl substituents had little impact on the efficiency and enantioselectivity of the difluoromethylation reactions (**28**–**31**, 92–97% e.e.). Substrates that bear internal and terminal alkenes, which could potentially react with aryl radicals, were effectively converted to the desired products (**32** and **33**, 90% and 85% e.e.). Alkyl chlorides, which are potentially labile to low-valent nickel intermediates, did not affect the difluoromethylation reaction (**34**, 94% e.e.). A modest decrease in enantioselectivity was observed when a phenyl group was adjacent to the alkyl iodide (**35**, 83% e.e.), probably due to the interaction of the phenyl group with the nickel centre. Substrates that contain imide (**36**), weak C–H bonds (**37**) or terminal alkyne (**38**) could also be converted to the desired products with good enantioselectivity (90–92% e.e.).

Next, the compatibility of various amide groups with this difluoromethylation protocol was evaluated (Fig. 4b). The presence of electron-donating, electron-neutral or electron-deficient substituents on phenyl groups had minimal impact on the enantioselectivity of the products (**39**–**44**, 91–98% e.e.). Functional groups such as ketone (**45**), ester (**46**) and carboxamide (**47**) were tolerated under the standard reaction conditions (94–96% e.e.). Notably, substrates that contain an aryl chloride or an aryl iodide, which could be problematic with low-valent Ni catalysis, were amenable to this protocol (**48** and **49**, 92%

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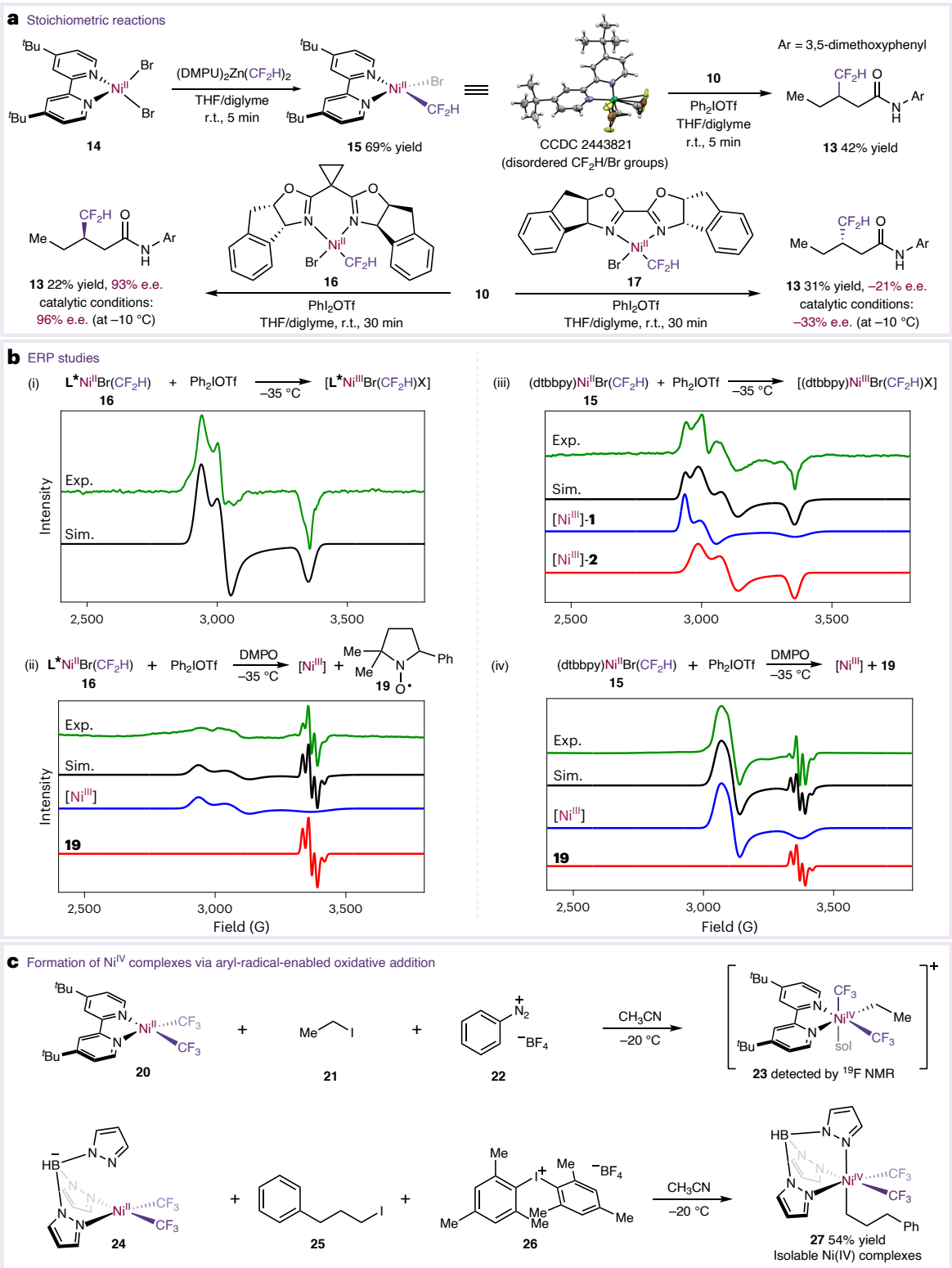

Fig. 3 | Evidence supporting the nickel(II)/(IV) cycle. a, Formation of the difluoromethyl product with the Ni^{II}–CF_2H species under diaryliodonium-enabled conditions. **b**, EPR studies on the involvement of paramagnetic Ni^{III} and aryl radical intermediates. EPR spectra were recorded at 77 K. **c**, Formation of alkyl–Ni^{IV} complexes via the aryl-radical-enabled oxidative addition of Ni^{II} complexes. ᵗBu, *tert*-butyl; exp., experimental EPR spectrum; sim., simulated EPR spectrum; Ph, phenyl.

and 96% e.e.), thereby enabling further diversification of the difluoromethylated products. Heterocycles, including fluorene (**50**), dibenzofuran (**51**) and thiophene (**52**), were also amenable to this protocol (92–95% e.e.). In addition to *N*-aryl amides, substrates derived from *N*-alkyl amides could also be converted to the desired products with high enantioselectivity (**53**–**55**, 85–91% e.e.). The compatibility with a diverse range of amide groups could enable the modular synthesis of chiral γ-difluoromethyl amines, as exemplified by the synthesis of

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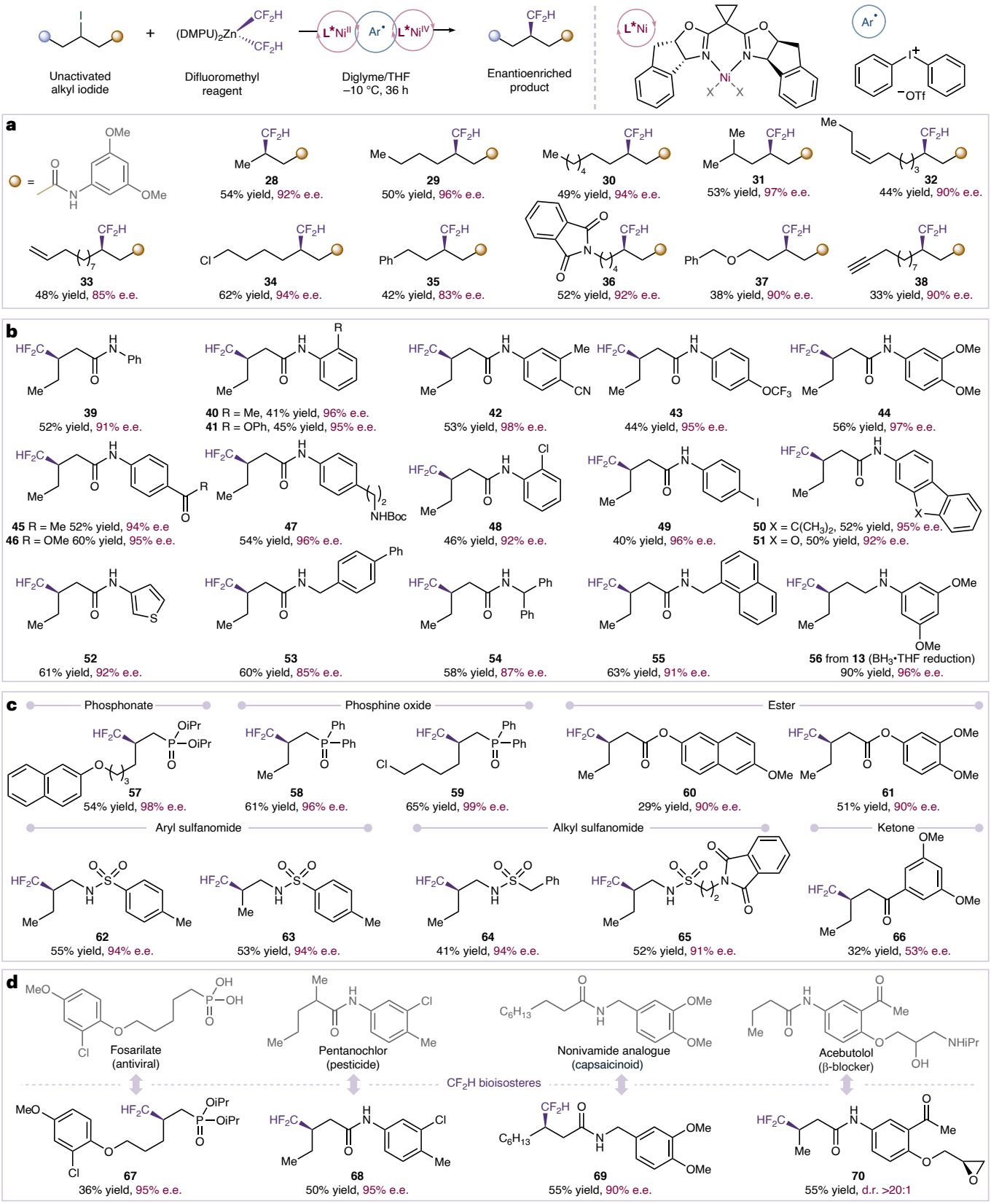

Fig. 4 | Scope of enantioconvergent difluoromethylation of alkyl iodides. Reactions were performed with 0.2 mmol alkyl iodide, 0.04 mmol of $NiBr_2$·DME and 0.06 mmol of **L*** in diglyme/THF (4/1) at −10 °C for 36 h. Yields were based on isolated products. **a**, Scope of alkyl groups. **b**, Scope of amide groups. **c**, Other classes of alkyl iodides. **d**, Synthesis of CF_2H analogues of bioactive molecules. Boc, *tert*-butyloxycarbonyl.

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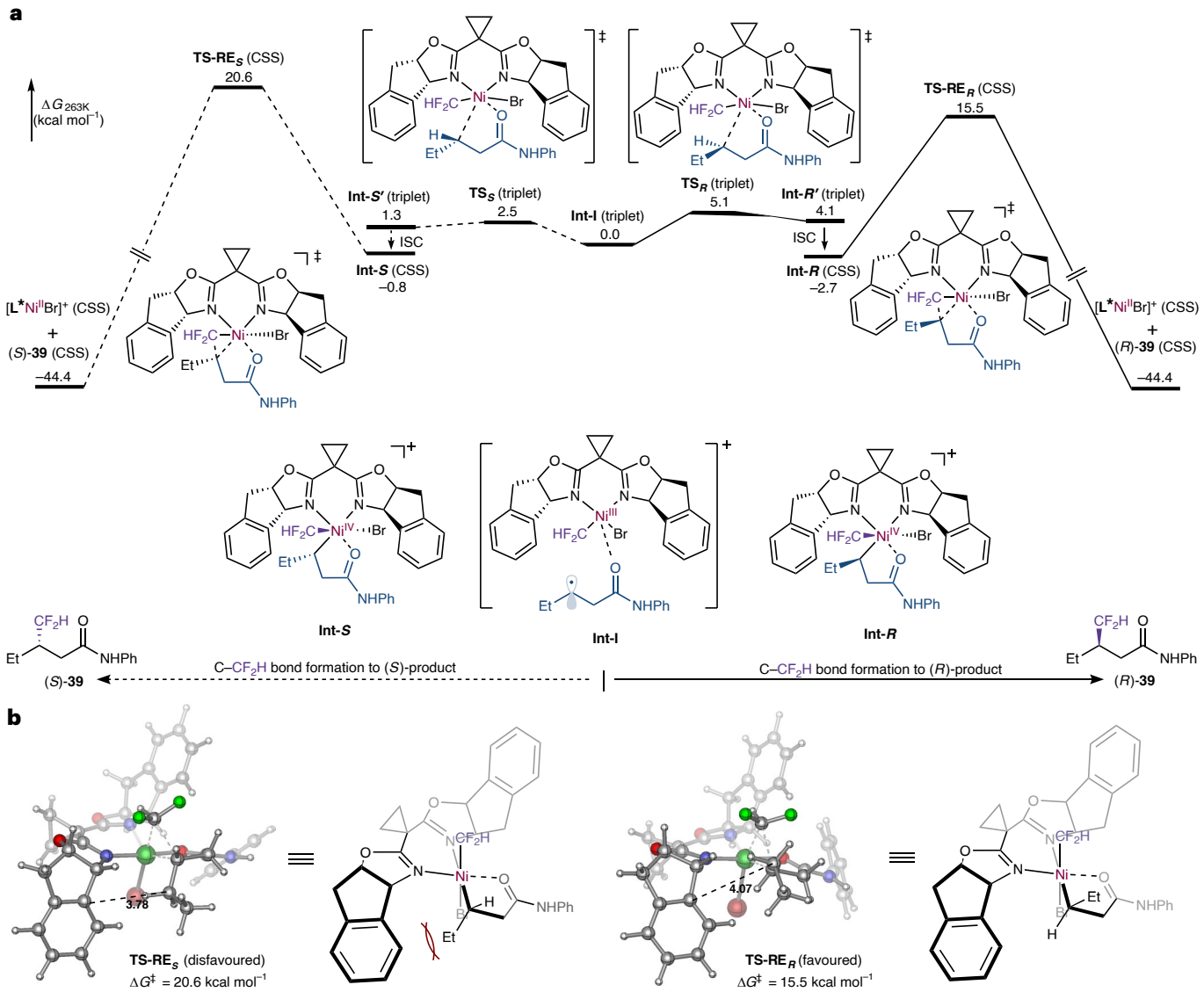

Fig. 5 | DFT calculations on the enantioselective difluoromethylation reaction. a, Computed reaction energy profile at the level of B3LYP-D3BJ/def2-TZVP// B3LYP-D3BJ/def2-SVP. The multiplicity of each stationary point is given in parentheses. **b**, Origin of the enantioselectivity of the difluoromethylation reaction.

56 via the borane reduction of **13** without the erosion of the chiral centre (90% yield, 96% e.e.).

With respect to other classes of weakly coordinating groups, we were pleased to find that substrates derived from phosphonates (**57**, 98% e.e.), phosphine oxides (**58** and **59**, 96% and 99% e.e.), esters (**60** and **61**, 90% e.e.), aryl sulfonamides (**62** and **63**, 94% e.e.) and alkyl sulfonamides (**64** and **65**, 94% and 91% e.e.) could undergo efficient asymmetric difluoromethylation, furnishing the desired CF_2H products in moderate-to-good yield and with high enantioselectivity (Fig. 4c). Meanwhile, a moderate yet promising enantioselectivity was observed for the difluoromethylation of a β-iodo ketone (**66**, 53% e.e.), probably due to the weaker coordination ability of the ketone group.

The synthetic utility of this enantioselective difluoromethylation protocol has been further highlighted by the synthesis of fluorinated analogues of biologically active molecules (Fig. 4d). For example, the CF_2H derivative of fosarilate—a phosphonate-based antiviral molecule—was obtained in synthetically useful yield and with high enantioselectivity (**67**, 95% e.e.). Likewise, the difluoromethylated analogues of pentanochlor, a pesticide, and nonivamide, a capsaicinoid, were successfully synthesized with excellent control of stereochemistry (**68** and **69**, 95% and 90% e.e.). Finally, the difluoromethylated analogue of the acebutolol precursor was accessed with good diastereoselectivity (**70**, diastereomeric ratio (d.r.) >20:1), with the sensitive epoxide group well tolerated under the mild reaction conditions.

Computational studies on the enantiodetermining step

Finally, we performed DFT calculations to understand the origins of the enantioselectivity of this Ni-catalysed difluoromethylation reaction, focusing on the reaction between the Ni^{III} intermediate and the alkyl radical (see Fig. 5a and Supplementary Figs. 24–26 for details). The computational studies indicate that the carbonyl group of the alkyl radical initially coordinates to the Ni^{III} intermediate **L***$Ni^{III}Br(CF_2H)$, forming a triplet catalyst–substrate complex **Int-I** (Fig. 5a), with a low Gibbs free energy barrier ($\Delta G^{\ddagger}$ = 4.9 kcal mol^{-1}; Supplementary Fig. 24). Subsequent Ni–C bond formation between the carbon-centred radical and the Ni centre, followed by the intersystem crossing (ISC), yields closed-shell singlet (CSS) Ni^{IV} intermediates, **Int-*R*** or **Int-*S***. These steps proceed with low free energy barriers ($\Delta G^{\ddagger}$ = 5.1 for $\mathbf{TS}_R$, $\Delta G^{\ddagger}$ = 2.5 for $\mathbf{TS}_S$). The resulting Ni^{IV} complexes undergo concerted C–CF_2H bond-forming reductive elimination to afford either (*R*)-**39** or (*S*)-**39** product. The transition state leading to the (*R*)-product, $\mathbf{TS\text{-}RE}_R$,

is lower in free energy ($\Delta G^{\ddagger}$ = 15.5 kcal mol^{-1}) than its (*S*)-counterpart, **TS-RE$_S$** ($\Delta G^{\ddagger}$ = 20.6 kcal mol^{-1}). The experimental value of e.e. of 91% for compound **39** corresponds to a 1.6 kcal mol^{-1} difference in the free energy barrier ($\Delta\Delta G^{\ddagger}$), which, although lower than the computed value of 5.1 kcal mol^{-1}, is consistent with the predicted stereochemical preference, favouring the formation of (*R*)-**39** as the major product. The discrepancy may arise from uncertainties in the ligand environment at the nickel centre. Furthermore, structural analysis of the enantiodetermining transition states shows that **TS-RE$_S$** has a higher energy barrier due to the steric repulsion between the ethyl group in the substrate and the indanyl group of the ligand (3.78 Å in **TS-RE$_S$** and 4.07 Å in **TS-RE$_R$**; Fig. 5b).

Conclusions

In summary, this work establishes an unconventional catalytic platform for asymmetric nickel catalysis by harnessing a Ni$^{II/IV}$ redox cycle, enabling enantioselective difluoromethylation of unactivated alkyl iodides with various weakly coordinating groups. This strategy addresses longstanding limitations associated with traditional Ni^{I}/Ni^{III}-based pathways, including their limited compatibility with electron-deficient nucleophiles and unactivated substrates. Mechanistic studies, including spectroscopic analysis and computational investigations, support an aryl-radical-enabled oxidative addition of Ni^{II} species and an enantiodetermining reductive elimination from Ni^{IV} intermediates. We anticipate that this catalytic platform will inspire the development of a broader class of asymmetric reactions leveraging high-valent nickel and other related high-valent metal intermediates.

Methods

Stoichiometric reactions of L*Ni^{II}Br(CF$_2$H) with alkyl iodides

In a nitrogen-filled glove box, an oven-dried 4-ml vial equipped with a magnetic stir bar was charged with NiBr$_2$·DME (15 mg, 0.05 mmol), **L*** (21 mg, 0.06 mmol) and tetrahydrofuran (THF) (0.5 ml). The mixture was stirred at room temperature (r.t.) for 30 min, after which (DMPU)$_2$Zn(CF$_2$H)$_2$ (9 mg, 0.02 mmol) was added. The resulting solution was stirred for an additional 5 min and was analysed by ^{19}F NMR with PhCF$_3$ as the internal standard. Subsequently, alkyl iodide **10** (18 mg, 0.05 mmol) and Ph$_2$IOTf (32 mg, 0.075 mmol; where OTf is trifluoromethanesulfonate) were added, and the reaction mixture was allowed to stir at r.t. for an additional 30 min. The crude reaction mixture was then analysed by ^{19}F NMR.

General procedure for EPR studies

All samples for EPR studies were prepared in a nitrogen-filled glove box. Reactions were carried out in 4:1 (v/v) THF:diglyme at −35 °C and quenched in liquid nitrogen before EPR analysis. EPR spectra were acquired on a Bruker EMXplus EPR X-band spectrometer equipped with a ColdEdge Stinger closed-cycle flow system. Measurements were performed in 4-mm quartz tubes with a rubber cap (Wilmad Lab glass, SKU: 707-SQ-250M) under the following conditions: temperature 77 K, modulation amplitude 4 G, microwave power 10 mW and microwave frequency ~9.46 GHz. The EPR spectra were simulated using EasySpin.

General procedure for enantioselective difluoromethylation

In a nitrogen-filled glove box, an oven-dried 4-ml vial equipped with a magnetic stir bar was charged with NiBr$_2$·DME (12.3 mg, 0.04 mmol, 20 mol%) and **L*** (21.4 mg, 0.06 mmol, 30 mol%), as well as 1.6 ml diglyme. The vial was closed with a polytetrafluoroethylene septum cap, and the solution was stirred at r.t. for 30 min. Then, iPr$_2$NEt (34.8 µl, 0.2 mmol, 1.0 equiv.) and Ph$_2$IOTf (154.9 mg, 0.36 mmol, 1.8 equiv.) were added to the vial. The reaction vial was sealed with parafilm and electrical tape, removed from the glove box and stirred for 10 min in a −10 °C cooling bath. A solution of (DMPU)$_2$Zn(CF$_2$H)$_2$ (127.2 mg, 0.3 mmol, 1.5 equiv.) and the alkyl iodide (0.2 mmol) in 0.4 ml THF was injected into the vial using a syringe. The reaction mixture was stirred at −10 °C for 36 h. After the completion of the reaction, the mixture was quenched with H$_2$O and extracted three times with Et$_2$O. The combined organic layer was washed with brine, dried over anhydrous MgSO$_4$ and concentrated in vacuo. The crude material was purified by column chromatography to afford the desired products.

Data availability

The data that support the findings of this study are available within the article and its Supplementary Information (experimental procedures, characterization data and DFT details) or from the corresponding authors upon request. The X-ray crystallographic coordinates for structures reported in this study have been deposited at the Cambridge Crystallographic Data Centre under deposition numbers 2443820 and 2443821. These data can be obtained free of charge from Crystallographic Data Centre at www.ccdc.cam.ac.uk/data_request/cif.

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Author contributions

X. Zhao and W.L. designed the experiments. X. Zhao., L.Y., C.W., X.W. and H.G. performed the synthetic experiments. A.T.P., X. Zhang., S.C.Y. and S.T. performed EPR experiments. C.-J.Y. and M.-J.C. conducted DFT calculations. J.A.K. performed the X-ray diffraction analysis. W.L. conceived and supervised the project and wrote this paper with contributions from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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